Decision Assistance and Steering in Health Insurance
SUPPLEMENTAL APPENDIX

Ian McCarthy & Evan Saltzman

August 2026

1 Constructing the Outside Option

We construct the outside option from five years of longitudinal Covered California data, treating a household as uninsured in the years in which it has no exchange coverage but remains eligible for the exchange market. For example, a household with exchange coverage in 2015 and 2016 is treated as uninsured, the outside option, in 2014, 2017, and 2018, provided it remained eligible for the exchange market in those years. A household may lose exchange-market eligibility if it gains access to employer-sponsored insurance or to public coverage such as Medicaid or Medicare. Our data do not record when a household loses eligibility, so we impute it from the Survey of Income and Program Participation (SIPP).

To impute consumer exchange market eligibility, we use data from the SIPP. The SIPP is well-suited for the imputation because (1) it asks the insurance status of respondents for every month over a three-year period (2013-2015) and (2) it includes detailed information on the chief reasons for consumers' coverage status, such as whether the respondent obtained or lost an offer of employer-sponsored insurance, moved in or out of California, or became eligible or ineligible for Medicare or Medicaid. For SIPP respondents who newly obtained or gave up individual market coverage, we construct a *transitioned* variable that indicates whether the respondent gained or lost eligibility for the individual market. The *transitioned* variable takes value 1 if the respondent (1) belongs to a household that obtained or lost an offer of employer-sponsored insurance; (2) moved out of or into California; (3) the respondent turned 65 and qualified for Medicare; and (4) the respondent became eligible for Medicaid following a drop in income. We estimate a logit model regression of the *transitioned* variable on the demographic variables available in both the SIPP and the Covered California data, including age, income, gender, race, and household size. We use the estimated logit to predict whether Covered California consumers observed for only some years of our study timeframe transitioned into or out of the exchange market. Consumers transitioning into or out of the exchange market are removed from our study population during years when they are not enrolled in an exchange plan.

2 Assistance Channels

The body groups formal decision assistance into two categories, agents and navigators. Covered California's enrollment records distinguish a finer set of service channels, which we map into those categories. Table A1 lists the underlying codes, a short description of each, the category we group it into, and its share of enrollment records over 2014–2019.

Table A1: Covered California Assistance Channels[a]

Code	Description	Grouped as	Share
CIA	Certified insurance agent	Agent	43.5%
PBE	Plan-based enroller	Agent	1.7%
SCR	Service-center representative	Navigator	12.2%
CEC	Certified enrollment counselor	Navigator	6.2%
CEW	Certified enrollment worker	Navigator	0.4%
Unassisted	Unassisted	Unassisted	36.1%

[a]Service-channel codes recorded in the Covered California enrollment data, 2014–2019, with each code's description, the higher-level category we group it into, and its share of individual enrollment records. Shares are of all enrollment records, assisted and unassisted.

The two commissioned or insurer-affiliated channels are the certified insurance agents and the plan-based enrollers, which together form the agent category. Certified insurance agents are independent private agents who hold appointments with a subset of insurers and are paid a commission by the insurer into which they enroll a consumer. They are the large majority of the agent category and the channel through which the commission steering we study operates. Plan-based enrollers are affiliated with a single health plan and enroll consumers into that insurer's products, so they are a small, structurally distinct group whose influence on insurer choice comes from that affiliation rather than from a commission on competing plans.

The navigator category pools the three non-commissioned channels, none of which is paid by any insurer. Certified enrollment counselors and workers are community-based assisters, and service-center representatives are Covered California's own telephone staff. We group these together because they share the absence of a commission incentive, though they differ in setting, and the service-center channel is the largest of the three.

We use the two-category grouping throughout the paper, because the commission mechanism we model operates at the level of whether a household was assisted by a commissioned agent rather than at the level of the specific channel. The one channel whose distinction bears on our interpretation is the plan-based enroller, since its influence on insurer choice reflects a single-insurer affiliation rather than a commission on competing plans. It is a small share of enrollments and is grouped with the certified agents in the assistance indicator used in estimation.

2.1 Excluding Plan-Based Enrollers

Plan-based enrollers are the one assistance channel that can place a consumer only with a single insurer, so their influence on insurer choice is a menu restriction rather than a response to commissions. We assess their effect on the demand estimates directly. Merging the raw service-channel code onto the estimation sample by household and year, plan-based enrollers account for 1.3 percent of the household-years in the 20 percent structural sample. Table A2 re-estimates the body demand specification with these households excluded, alongside the full-sample estimates. Dropping them leaves every coefficient essentially unchanged. The commission-by-broker coefficient, which carries the steering mechanism, is 0.021 in both columns, and no coefficient moves by more than about 0.02. The single-insurer channel does not drive the estimated steering.

Table A2: Demand Estimates Excluding Plan-Based Enrollers[a]

	Full sample	Excl. PBE
Premium	-0.682	-0.673
Navigator × silver	1.381	1.371
Broker × silver	1.427	1.431
Navigator × bronze	1.812	1.790
Broker × bronze	1.926	1.910
Navigator × premium	0.462	0.456
Broker × premium	0.487	0.477
Commission × broker	0.021	0.021
λ (nesting parameter)	0.674	0.667

[a]Body demand specification re-estimated on the 20% structural sample with plan-based-enroller households excluded, alongside the full-sample estimates. Plan-based enrollers are identified by merging the raw Covered California service-channel code onto the estimation households by household and year, and account for 1.3% of household-years in the sample. Premiums are net of subsidy and measured in hundreds of dollars.

2.2 Commission Utility and Menu Restriction

The certified insurance agents present the full menu but may still steer a household within it, short of the plan-based enroller's single-insurer restriction. We represent that steering through the commission coefficient in utility. To check that this does not distort the plan choices the model predicts, we simulate a gatekeeper process and confirm that our specification reproduces its choices.

The simulation draws a market of six insurers whose commissions vary, and generates choices from a process with no commission term in utility. Unassisted households choose their preferred plan from the full menu. Broker households instead face a restricted menu, in which each insurer appears with a probability that rises in its commission, and choose their preferred plan from what is presented. This produces the steering the commissions are meant to induce, with broker households moving away from the low-commission insurers and toward the high-commission ones, entirely through availability rather than preference.

We then fit our commission-in-utility model to these choices, with the full menu available to every household and the commission entering utility for broker households. Table A3 reports the resulting shares. The fitted model recovers a positive commission coefficient and reproduces the gatekeeper broker shares to within about one percentage point, with a correlation of 0.9995. A menu restriction and a commission in utility are observationally equivalent for the plan choices the model predicts, and our specification loads the restriction onto the commission coefficient without distorting those predictions.

This equivalence is a statement about the positive predictions and the demand fit, not about the counterfactuals. A menu restriction fixed by an agent's appointments would not respond to a commission change the way a utility term does, so the two models diverge once commissions are altered. We attribute the steering to the commission margin in the counterfactuals, as the body describes, and read this exercise as evidence that the simplification does not distort the estimated choice behavior.

Table A3: Menu Restriction and the Commission Coefficient[a]

Insurer	Commission	Unassisted	Broker (gatekeeper)	Broker (fitted)
1	0.5	0.061	0.022	0.025
2	1.0	0.058	0.031	0.030
3	1.5	0.010	0.008	0.007
4	2.5	0.267	0.235	0.225
5	3.5	0.473	0.500	0.511
6	4.5	0.131	0.205	0.202

[a]Simulated plan-choice shares. Choices are generated by a gatekeeper process in which broker households face a menu that includes each insurer with probability rising in its commission, with no commission term in utility. The commission-in-utility model is then fit on the full menu. Columns report the unassisted shares, the gatekeeper broker shares, and the broker shares predicted by the fitted commission model. The fitted commission coefficient is positive, and the fitted broker shares match the gatekeeper shares to within about one percentage point, with a correlation of 0.9995.

3 Additional Design-Based Results

The design-based estimates in the body of the paper come from a prediction-based estimator. We fit the outcome model on unassisted households, weighted by inverse propensity weights, predict the counterfactual outcome for assisted households, and form the ATT as the difference between what assisted households chose and what the model predicts they would have chosen. Assistance never appears as a regressor, so there is no assistance coefficient to report and no control function anywhere in the estimation. This appendix reports the complementary pooled specifications, in which assistance enters as a covariate, so its coefficient can be reported across a sequence of control sets.

3.1 Dominated Choices

Table A4 reports linear probability models of the dominated-choice indicator on the sample of households with at least one dominated plan in the choice set. Column (1) is unconditional. Column (2) adds the household demographics that enter the propensity score, namely age composition, gender, race and ethnicity, and household size. Income does not appear, because the sample is defined by cost-sharing-reduction eligibility and therefore contains only households below 200% of the federal poverty level, leaving the income brackets used elsewhere in the paper with no variation to exploit. Column (3) adds a new-enrollee indicator and year fixed effects. Columns (4) and (5) are the two terminal specifications. Column (4) adds the broker-density control function without region fixed effects, and column (5) adds region fixed effects without the control function. Standard errors are clustered by rating region throughout, and all specifications are inverse-propensity weighted.

The assistance coefficient is stable across columns (1) through (3) and (5), moving only from -0.0152 to -0.0168 as controls accumulate and settling at -0.0161 once region fixed effects absorb differences in market composition. Column (4) is the exception. With the control function included, the assistance coefficient falls to -0.2564 and the control function residual enters at $+0.2402$, so that the two nearly offset and their sum, -0.0162, reproduces the estimate in column (3). The two offset because the first stage is too weak to separate selection from the treatment effect, so the column is a reparameterization rather than independent evidence.

The two terminal columns never appear together, and not because the combination is underpowered. Broker density varies only across rating region and year, so region fixed effects leave the instrument with no variation to exploit. Within a region the number of licensed agents is close to a deterministic function of the year, which means a region-specific control function is identically zero rather than weak. Clustering standard errors at the level at which the instrument actually varies, the partial F statistic on broker density is 3.6 and the incremental R^2 is 0.001, against a partial F of roughly 7,600 when the clustering is ignored. The apparent strength of the instrument in the unclustered specification reflects the number of households rather than the number of markets. The instrument's first-stage relationship with assistance take-up is positive and in the expected direction, but it rests on too few markets for meaningful inference. Nonetheless, even with a weak first stage, the pooled results are consistent with the body estimates. The implied total effect in column (4) matches the stable columns, and every specification in the table points to the same modest reduction in dominated choices.

Table A5 reports the body estimator re-run on new enrollees alongside the all-enrollee estimates. The body sample is all enrollees so that the design-based analysis and the structural model are estimated on the same households. Restricting to new enrollees produces larger effects for every channel, consistent with assistance mattering most for households making a first plan choice, and leaves the ordering across channels unchanged.

Table A4: Dominated Choices, Pooled Specifications[a]

Variable	(1)	(2)	(3)	(4)	(5)
Assisted	-0.0152***	-0.0152***	-0.0168***	-0.2564**	-0.0161***
	(0.0012)	(0.0011)	(0.0011)	(0.1181)	(0.0007)
CF Residual				0.2402*	
				(0.1185)	
HH Size		-0.0018***	-0.0007	0.0120*	-0.0003
		(0.0004)	(0.0005)	(0.0064)	(0.0005)
New Enrollee			0.0168***	0.0167***	0.0168***
			(0.0027)	(0.0027)	(0.0027)
Demographics		X	X	X	X
Year FE			X	X	X
Region FE					X
Control Function				X	
Observations	2,993,294	2,993,294	2,993,294	2,993,294	2,993,294
R^2	0.002	0.003	0.012	0.013	0.016

[a]Linear probability models of an indicator for selecting a dominated plan, on the sample of households with at least one dominated plan in the choice set. All specifications are weighted by inverse propensity weights, with standard errors clustered by rating region. Demographics include age composition, gender, race and ethnicity, and household size. Columns (4) and (5) are the two terminal specifications described in the text and are never combined. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A5: Dominated Choice ATT, All Enrollees Versus New Enrollees[a]

Channel	All Enrollees	New Enrollees
Any Assistance	-0.0156	-0.0226
Agent/Broker	-0.0183	-0.0252
Navigator	-0.0090	-0.0180

[a]Prediction-based ATT of decision assistance on the probability of selecting a dominated plan, weighted by inverse propensity weights. The all-enrollee column is the body estimate. The new-enrollee column re-runs the same estimator on households enrolling for the first time.

3.2 Plan Choice

Table A6 reports the companion sequence for plan choice. The model is a conditional logit over the plans in each household's choice set, pooled across assisted and unassisted households and weighted by inverse propensity weights. Assistance enters through its interaction with metal tier, separately by channel, using the same navigator and broker split as the structural model in the main paper. Column (1) includes plan attributes only. Column (2) adds the premium-by-demographic interactions that constitute the full control set in the body specification. Column (3) adds the broker-density control function, again interacted with plan attributes.

Three features of this model differ from the dominated-choice table. First, there is no region fixed effects column. A household-level region indicator is constant across the alternatives in that household's choice set and is differenced out of the conditional logit, so it does not control for anything. The term that would actually absorb region here is region interacted with metal tier and insurer, which adds well over a hundred parameters and leaves the insurer block empty wherever an insurer does not operate in a region. We do not estimate it, and region is therefore uncontrolled in this model rather than absorbed.

Second, assistance interacts only with silver and bronze. Catastrophic plans are excluded from the sample, so the four metal tiers exhaust the alternatives and a full set of four assistance-by-metal interactions sums to a household constant, which is differenced out. Gold and platinum share the omitted category, matching the base specification.

Third, the control function terms enter as interactions with plan attributes for the same reason. The first-stage residual is a household-level quantity and cannot enter a conditional logit on its own.

Column (3) shows the same pattern as the control-function column of the dominated-choice table. The assistance-by-silver interactions inflate and the control-function-by-silver term moves in the opposite direction, consistent with the weak first stage documented above, and we read the column as a reparameterization of column (2) rather than independent evidence. The steering pattern itself, toward silver and away from bronze with brokers steering more strongly than navigators, is stable across all three columns.

Table A6: Plan Choice, Pooled Specifications[a]

Variable	(1)	(2)	(3)
Premium	-0.9563 (0.0021)	-0.5257 (0.0043)	-0.4857 (0.0044)
Navigator × Silver	0.2960 (0.0093)	0.2829 (0.0094)	1.8499 (0.0268)
Navigator × Bronze	-0.3046 (0.0104)	-0.3311 (0.0105)	-0.3471 (0.0343)
Broker × Silver	0.6039 (0.0077)	0.6007 (0.0078)	2.0867 (0.0252)
Broker × Bronze	0.0562 (0.0084)	0.0581 (0.0086)	0.0239 (0.0328)
CF × Silver			-1.5822 (0.0263)
CF × Bronze			0.0008 (0.0339)
Premium × demographics		X	X
Control function			X
Households	1,067,827	1,067,827	1,067,827
Log-likelihood	-2,668,062	-2,619,902	-2,608,811

[a]Conditional logit over the plans in each household's choice set, pooled across assisted and unassisted households and weighted by inverse propensity weights. Premiums are net of subsidy and measured in hundreds of dollars. Plan attributes are metal tier, network type, health savings account eligibility, and insurer indicators. CF denotes the broker-density control function interacted with the indicated plan attribute. Standard errors in parentheses.

4 Choice Probabilities and Pricing Formulas

This section collects the derivatives that underlie the demand model and the insurer first-order conditions in the main paper, written in the notation used there. Household i in market m and year t chooses among the exchange plans $j \in J_{mt}$ and the outside option, with deterministic utilities V_{ijt} and $V_{i0t} = \alpha_{it}\,\rho_{it}$, nesting parameter λ, and household price coefficient α_{it}.

Choice probabilities. Let $I_{it} = \log \sum_{k \in J_{mt}} e^{V_{ikt}/\lambda}$ denote the inclusive value of the exchange nest. The within-nest and nest probabilities are

$$s^g_{ijt} = \frac{e^{V_{ijt}/\lambda}}{\sum_{k \in J_{mt}} e^{V_{ikt}/\lambda}}, \qquad s^{\text{nest}}_{it} = \frac{e^{\lambda I_{it}}}{e^{\lambda I_{it}} + e^{V_{i0t}}}, \tag{1}$$

and the plan choice probability is $q_{ijt} = s^g_{ijt}\, s^{\text{nest}}_{it}$, which is the nested-logit expression in the main text. Market shares aggregate the household probabilities by size weight, $s_{jt} = \sum_i w_i\, q_{ijt}$.

Price derivatives. The price coefficient enters utility linearly, so $\partial V_{ijt}/\partial p_{ijt} = \alpha_{it}$. Differentiating the choice probability with respect to the utility of an arbitrary plan l gives

$$\frac{\partial q_{ijt}}{\partial V_{ilt}} = q_{ijt}\left[\frac{\mathbf{1}\{j = l\}}{\lambda} + \frac{\lambda - 1}{\lambda}\, s^g_{ilt} - q_{ilt}\right]. \tag{2}$$

For a plan other than the benchmark, a change in its base premium moves only its own consumer premium, so $\partial q_{ijt}/\partial p_{lt} = \alpha_{it}\, \partial q_{ijt}/\partial V_{ilt}$. For the benchmark plan the subsidy chain rule applies, since raising the benchmark premium raises the subsidy available to every plan and so lowers the consumer premium of all competing plans while leaving the benchmark's own consumer premium unchanged for subsidized households. The market-share derivatives $\partial s_{jt}/\partial p_{lt}$ follow by aggregating Equation (2) across households.

Pricing first-order condition. Stacking the plan-level first-order conditions in the main paper into matrix form yields the equilibrium markup $p - mc = \Omega^{-1}\big(s + \Omega^B \mathcal{C}\big)$, where p, mc, s, and $\mathcal{C}$ are the vectors of base premiums, marginal costs (net of risk adjustment and reinsurance), market shares, and commissions, the ownership matrix has elements $\Omega_{jl} =$

$-\mathbf{1}\{f(j) = f(l)\}\, \partial s_l / \partial p_j$, and Ω^B is the analogous matrix built from broker-assisted demand derivatives only. Marginal cost is net of the risk-adjustment transfer, whose price sensitivity enters the condition through

$$\frac{\partial RA_k}{\partial s_m} = \left(-\frac{r_k\, r_m}{S_r^2} + \frac{u_k\, u_m}{S_u^2} \right) \bar{p}_t, \qquad \frac{\partial RA_k}{\partial p_l} = \sum_m \frac{\partial RA_k}{\partial s_m} \frac{\partial s_m}{\partial p_l}, \tag{3}$$

where r_k is plan k's risk score, u_k its actuarial value scaled by a moral-hazard factor, $S_r = \sum_k r_k s_k$ and $S_u = \sum_k u_k s_k$ are the enrollment-weighted risk and utilization totals, and $\bar{p}_t$ is the market average premium. Evaluating the first-order condition at observed premiums and shares and solving for mc recovers plan-level marginal cost, the inversion described in the main paper.

Commission first-order condition. Differentiating the same variable profit with respect to insurer f's commission scale k_f gives the commission first-order condition in the main paper. Its left-hand side is the marginal benefit of a higher commission, the additional variable profit from the broker-assisted enrollment the commission attracts, which operates through the steering coefficient β^C and the derivatives $\partial q^B_{jt}/\partial k_f$, together with the induced change in risk-adjustment transfers. Its right-hand side is the marginal outlay $\sum_j w_{jt}\, q^B_{jt}$ scaled by the per-insurer wedge μ_f. Calibrating μ_f so that the observed schedule satisfies this condition makes the observed premiums and commissions a joint fixed point, the commission-side counterpart to recovering marginal cost from the pricing condition.

5 Demand Specification Sensitivity

The main paper reports a single demand specification, the nested logit whose estimates appear there. This section reports how those estimates change as we add terms to the utility function and how they respond to the premium control function. Table A7 reports four specifications, each estimated on the same 20% sample and the same 114 rating-region-by-year cells as the body model. Column (1) is a nested logit with plan attributes and a single premium coefficient. Column (2) adds the demographic heterogeneity in the price coefficient and in metal preference. Column (3) adds the assistance and commission steering terms and is the body specification, reproducing the main-text demand estimates exactly. Column (4) adds the broker-density control function.

We summarize price sensitivity by the enrollment-weighted mean own-price elasticity rather than by the premium coefficient. The premium coefficient and the nesting parameter are jointly identified and trade off against one another as terms are added, so neither is stable on its own. The nesting parameter rises from 0.38 to 0.67 across the columns and the premium coefficient more than doubles in magnitude, but the own-price elasticity they jointly imply stays between -0.75 and -0.82. The elasticity is measured with respect to the net premium households pay. It is inelastic because most enrollees are heavily subsidized and face little premium at the margin.

We likewise report the assistance results as marginal effects on plan choice rather than as log-odds coefficients. Assigning navigator assistance to every household raises the share choosing a silver plan by 7.5 percentage points, and broker assistance raises it by 7.1 percentage points, against an unassisted silver share near one half. The two effects are close even though the broker metal coefficient is larger, because brokers are also less price-sensitive, which offsets part of the stronger steering toward silver. The commission-by-broker coefficient is 0.02.

Columns (3) and (4) differ only in the broker-density control function. Adding it leaves every quantity nearly unchanged, including the own-price elasticity, the nesting parameter, and both assistance effects. The instrument is too weak at the level of variation that matters to separate the control-function residual from the premium coefficient, as the plan-choice results in Section 3 document, so the two columns are a reparameterization rather than independent evidence. Saltzman (2019) similarly finds that this correction leaves the nesting parameter essentially unchanged, with only a small downward bias in premium sensitivity when it is omitted.

Table A7: Demand Specification Sensitivity[a]

	(1)	(2)	(3)	(4)
Mean own-price elasticity	-0.75	-0.82	-0.78	-0.78
λ (nesting parameter)	0.38	0.45	0.67	0.67
Navigator effect on silver (pp)			7.5	7.6
Broker effect on silver (pp)			7.1	7.1
Commission $\times$ broker			0.021	0.021

[a]Four nested-logit demand specifications, each estimated on the 20% sample and the 114 rating-region-by-year cells used in the body. Column (1) is a nested logit with plan attributes and a single premium coefficient; column (2) adds demographic heterogeneity in the price coefficient and in metal preference; column (3) adds the assistance and commission steering terms and reproduces the main-text demand estimates; column (4) adds the broker-density control function. The mean own-price elasticity is the enrollment-weighted average with respect to the net premium; the navigator and broker effects are the average marginal effect of assistance on the probability of choosing a silver plan, in percentage points. Empty cells denote terms not included in that specification.

6 Cost Sharing and the Objective Welfare Measure

The objective welfare measure in the main paper values each insured plan at the negative of the household's annual premium plus the mean-variance certainty equivalent of its out-of-pocket exposure. This section states the cost-sharing schedule and the out-of-pocket calculation behind that measure.

Because Covered California standardizes its benefit designs, a plan's cost-sharing is fixed by its metal tier and cost-sharing-reduction level rather than estimated. Table A8 reports the 2019 schedule. The full 2014–2019 schedule, which the welfare computation uses, follows the same tier structure, with deductibles and out-of-pocket maxima that rose over the period.

Table A8: Covered California Standardized Cost Sharing, 2019[a]

Tier	AV	Deductible	Coins.	MOOP
Bronze	0.60	\$6,300	100%	\$7,550
Bronze (HSA)	0.60	\$6,000	40%	\$6,650
Silver	0.70	\$2,500	20%	\$7,550
Silver (CSR 73)	0.73	\$2,200	20%	\$6,300
Silver (CSR 87)	0.87	\$650	15%	\$2,600
Silver (CSR 94)	0.94	\$75	10%	\$1,000
Gold	0.80	\$0	20%	\$7,200
Platinum	0.90	\$0	10%	\$3,350

[a]Standardized patient-centered benefit designs for the 2019 plan year. AV is the statutory actuarial value; the deductible and maximum out-of-pocket (MOOP) are annual individual amounts; coinsurance is the enrollee share of spending above the deductible. CSR denotes the cost-sharing-reduction silver variants available to enrollees below 250% of the federal poverty level, and the bronze health-savings-account variant is shown separately. The full 2014–2019 schedule follows the same tier structure and is used in the welfare computation.

For a plan with deductible d, coinsurance rate c, and maximum out-of-pocket M, out-of-pocket spending at realized annual medical spending s is

$$o(s) = \min\{\min(s,\, d) + c\,\max(s - d,\, 0),\; M\}\,. \tag{4}$$

The household pays all spending up to the deductible, the coinsurance share of spending above it, and never more than the maximum out-of-pocket.

We treat annual individual spending as lognormal with mean $\bar{s}$ and coefficient of variation ν. Writing $\sigma^2 = \log(1+\nu^2)$, the underlying normal has variance σ^2 and mean $\log\bar{s} - \sigma^2/2$, so the spending distribution has mean $\bar{s}$ by construction. We integrate the schedule in Equation (4) against this distribution by deterministic quadrature over 400 evenly spaced midpoint quantiles, which fixes the mean and variance of out-of-pocket spending, $\mathbb{E}[o]$ and $\mathrm{Var}[o]$, without simulation noise. The plan's objective value is then

$$V^{obj} = -\left(P + \mathbb{E}[o] + \tfrac{\rho}{2}\,\mathrm{Var}[o]\right), \tag{5}$$

with P the annual premium and ρ the coefficient of absolute risk aversion. The variance term gives cost-sharing depth weight beyond its effect on expected out-of-pocket spending, so a household will pay more than the actuarial difference to move from a bronze plan to a platinum one.

The mean $\bar{s}$ is household-specific, set from Medical Expenditure Panel Survey total-expenditure cell means by age composition and income bracket, so that expected spending follows the household rather than the plan and the measure carries no moral-hazard response. The coefficient of variation ν and the risk-aversion coefficient ρ are the two calibrated inputs, set to 2.5 and 2.3×10^{-4} per dollar as described in the main paper, the latter the marketplace benchmark in Handel (2013). The uninsured option is valued separately, since it carries no plan cost-sharing, using the realized out-of-pocket and social-cost construction described in the main text.

References

Handel, Benjamin R., "Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts," *American Economic Review*, December 2013, *103* (7), 2643–2682.

Saltzman, Evan, "Demand for health insurance: Evidence from the California and Washington ACA exchanges," *Journal of health economics*, 2019, *63*, 197–222.